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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

np.random.seed(42)
raw_data = {
    'Quantity': np.random.randint(-10, 100, 400),
    'UnitPrice': np.random.uniform(0.5, 50.0, 400),
    'Country': np.random.choice(['United Kingdom', 'Germany', 'France', 'EIRE', 'Spain'])
    'Target': np.random.choice([0, 1], 400)
}
df = pd.DataFrame(raw_data)
df.loc[df.sample(frac=0.05).index, 'Quantity'] = np.nan

df['Quantity'] = df['Quantity'].fillna(df['Quantity'].median())

encoder = LabelEncoder()
df['Country'] = encoder.fit_transform(df['Country'])

plt.figure(figsize=(8, 5))
df['Country'].value_counts().plot(kind='bar', color='skyblue')
plt.title('Customer Distribution Graph')
plt.xlabel('Country (Encoded)')
plt.ylabel('Count')
plt.tight_layout()
plt.show()

X = df[['Quantity', 'UnitPrice', 'Country']]
y = df['Target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model_lr = LogisticRegression()
model_dt = DecisionTreeClassifier()
model_knn = KNeighborsClassifier()

model_lr.fit(X_train, y_train)
model_dt.fit(X_train, y_train)
model_knn.fit(X_train, y_train)

pred_lr = model_lr.predict(X_test)
pred_dt = model_dt.predict(X_test)
pred_knn = model_knn.predict(X_test)

acc_lr = accuracy_score(y_test, pred_lr)
acc_dt = accuracy_score(y_test, pred_dt)
acc_knn = accuracy_score(y_test, pred_knn)

cm_lr = confusion_matrix(y_test, pred_lr)
cm_dt = confusion_matrix(y_test, pred_dt)
cm_knn = confusion_matrix(y_test, pred_knn)

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```
print("Logistic Regression Accuracy:", acc_lr)
print("Logistic Regression CM:\n", cm_lr)
print("-" * 30)
print("Decision Tree Accuracy:", acc_dt)
print("Decision Tree CM:\n", cm_dt)
print("-" * 30)
print("KNN Accuracy:", acc_knn)
print("KNN CM:\n", cm_knn)
print("-" * 30)

models = ['Logistic Regression', 'Decision Tree', 'KNN']
accuracies = [acc_lr, acc_dt, acc_knn]

plt.figure(figsize=(8, 5))
plt.bar(models, accuracies, color=['crimson', 'teal', 'darkorange'])
plt.title('Model Accuracy Comparison Graph')
plt.xlabel('Machine Learning Models')
plt.ylabel('Accuracy Score')
plt.ylim(0, 1.0)
plt.tight_layout()
plt.show()
```

